

Lab 3: Architecture Justification

Architecture Selection: "We chose Layered Architecture for the Self-Service Coffee Kiosk System."

Architectural Choice:

We selected the Layered Architecture style, specifically dividing the system into Presentation, Business, and Data layers.

Two Reasons:

1. Separation of Concerns: The kiosk involves hardware interactions (touch screen, receipt printer) and business rules (pricing, menu). Layered architecture isolates the UI from business logic, making it easier to update the touch interface without affecting order processing.
2. Simplicity and Maintainability: For a standalone kiosk, layered architecture is straightforward to implement and maintain. It perfectly aligns with the kiosk's cohesive functionality (menu display -> ordering -> payment -> receipt).

Security Advantage:

The layered approach enhances security by isolating the Payment Service Component within the Business Layer. Direct access from the User Interface to the Database or Payment Service is restricted, ensuring that sensitive credit card transactions and data are protected from frontend vulnerabilities.

Performance Benefit:

By organizing functionality into distinct layers, communication pathways are direct and predictable (UI to Order Manager, Order Manager to Payment/Printer/DB). This local in-memory communication within a standalone kiosk avoids the network latency that would be present in distributed architectures (like Microservices), resulting in fast, responsive touch interactions for the customer.